

Question Bank:

Class-2

- 1) Find the finite Fourier Cosine transform of $f(t) = t$, if the function is defined in the interval $[0, \pi]$. Solution: $C_0 = \frac{\pi^2}{2}$, $C_n = \frac{\cos n\pi - 1}{n^2}$.

- 2) Find the Fourier Cosine transform of $f(t) = \sinh(at)$, $a > 0$ if the function is defined on $[0, \pi]$. Also find the inverse finite Fourier Cosine transform.

Solution: $C_0 = \frac{\cosh a\pi - 1}{a}$, $C_n = \frac{a}{a^2 + n^2} [\cosh a\pi \cos n\pi - 1]$ and the Inverse Finite

Fourier Cosine transform is $f(t) = \frac{\cosh a\pi - 1}{\pi a} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{a}{a^2 + n^2} [\cosh a\pi \cos n\pi - 1] \cos nt$.

- 3) Find the Finite Fourier cosine transform of $f(t) = e^{-t}$, if the function is defined on $[0, l]$. Solution: $C_0 = 1 - e^{-l}$, $C_n = \frac{1}{1 + n^2} [1 - e^{-l} \cos nl]$.

- 4) Find the Fourier sine transform of $f(t) = e^{-t}$, if the function is defined on $[0, \pi]$.

Solution: $S_n = \frac{n}{1 + n^2} [1 - e^{-\pi} \cos n\pi]$.

- 5) Find the Fourier sine transform of $f(t) = \sin(at)$, $a > 0$, if the function is defined on $[0, \pi]$. Also find the inverse finite Fourier Sine transform.

Solution: $S_n = \frac{1}{2} \left[\frac{\sin(n-a)\pi}{n-a} - \frac{\sin(n+a)\pi}{n+a} \right]$ and the inverse finite

Fourier Cosine Transform is $f(t) = \frac{1}{\pi} \sum_{n=1}^{\infty} \left[\frac{\sin(n-a)\pi}{n-a} - \frac{\sin(n+a)\pi}{n+a} \right] \sin nt$.

- 6) Find the Fourier sine transform of $f(t) = \cosh(bt)$, if the function is defined on $[0, l]$

Solution: $S_n = \frac{-n}{n^2 + b^2} [\cos nl \sinh nl + 1]$.